Neural Quantum States for Many-Body Ground States and Real-Time Dynamics

Cross-method benchmarks | 2D NQS advantage | Bose-Hubbard crossover | TDVP quench | Hybrid pretraining

VMC SR TDVP DMRG autoregressive NQS sector projection

Motivation. Neural quantum states (NQS) provide a flexible variational framework for many-body systems. They encode correlations through network weights without imposing a 1D tensor-network factorisation, which becomes especially useful in two dimensions and in real-time dynamics where matrix-product-state efficiency erodes. This highlight condenses six validated summaries into one portfolio: cross-method benchmarking, 2D TFIM scaling, Bose-Hubbard sector projection, TDVP quench dynamics, hybrid pretraining, and from-scratch time-dependent NQS.

Ground-state workstream

1D Heisenberg benchmarking establishes an entanglement bridge across VQCs, RBMs, and DMRG. The same validation chain then moves to 2D TFIM, where NQS surpass DMRG at lower parameter count, and to the Bose-Hubbard model where U(1) sector projection captures the Mott-superfluid crossover with high accuracy.

Dynamics workstream

Exact-diagonalisation-validated TFIM quenches expose what translation symmetry, then Z2 parity, actually fix in NetKet TDVP. A hybrid snapshot-to-VMC pipeline provides a reliable warm start, and a separate PyTorch t-NQS implementation maps the fidelity-conservation Pareto frontier for interval training.

Progressive validation chain

Every production result is grounded in a reference ladder: exact checks where available, then controlled variational scaling, then larger-system transfer. This is what makes the performance claims defensible rather than anecdotal.

4.3x	<0.06%	91%	1.91x
NQS param. advantage (6x6 2D)	Bose-Hubbard energy error	Z2 spurious m_z reduction	Hybrid VMC speedup

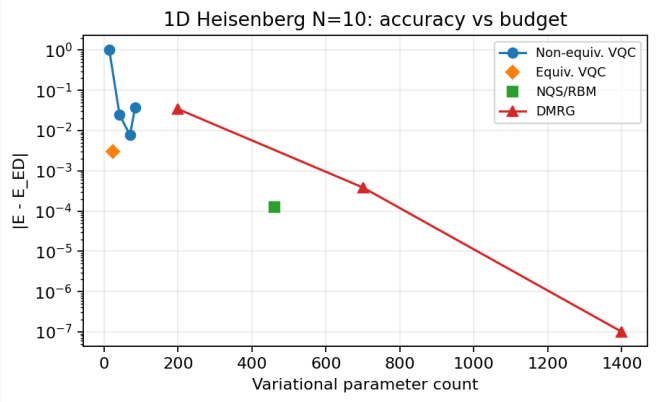
Ground-State Benchmarks: Cross-Method Comparison and 2D NQS Advantage

Heisenberg N=10 | TFIM 6x6 | accuracy-vs-budget figures

1D Heisenberg chain (N=10, PBC)

The portfolio starts with a five-way benchmark on the antiferromagnetic Heisenberg chain. DMRG is strongest in 1D, NQS reaches near-exact accuracy at moderate cost, and the equivariant VQC is clearly more parameter-efficient than the non-equivariant circuit at matched accuracy. The half-chain entanglement diagnostic explains the hierarchy: the NQS nearly matches the exact entropy, while the shallow circuits retain a persistent entanglement deficit.

Representative accuracy-vs-budget plot



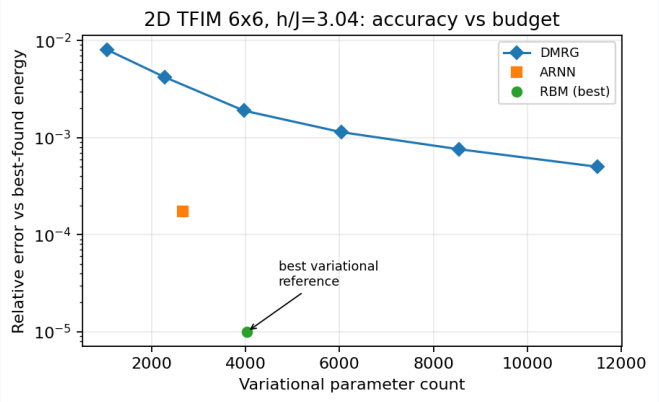
Representative points from the validated summary: DMRG wins asymptotically in 1D, NQS reaches 1.24e-4 error at 460 parameters, and symmetry improves VQC efficiency at low parameter count.

Method	Parameters	|E - E_ED|	S_N/2
DMRG (chi=32)	~1,400	<= 1e-7	1.1241
NQS/RBM (alpha=2)	460	1.24e-4	1.1252
Equiv. VQC (12L)	24	3.07e-3	1.009
Non-equiv. VQC (6L)	42	2.46e-2	0.992

2D TFIM 6x6 near criticality (h/J = 3.04)

At 6x6 sites, exact diagonalisation is inaccessible. The benchmark therefore uses the best variational energy found in the sweep as an internal reference. The ARNN reaches relative error 1.74e-4 with 2664 parameters, while DMRG at chi=14 needs 11480 parameters and still sits at 5.02e-4. This is the clearest ground-state scaling result in the portfolio: a 4.3x parameter-efficiency advantage for NQS in 2D.

6x6 TFIM scaling plot



DMRG improves only gradually across the tested chi sweep, while both NQS points sit well below the DMRG curve. The RBM defines the best-found variational reference; the ARNN beats DMRG chi=14 with 4.3x fewer parameters.

Bose-Hubbard with Sector Projection

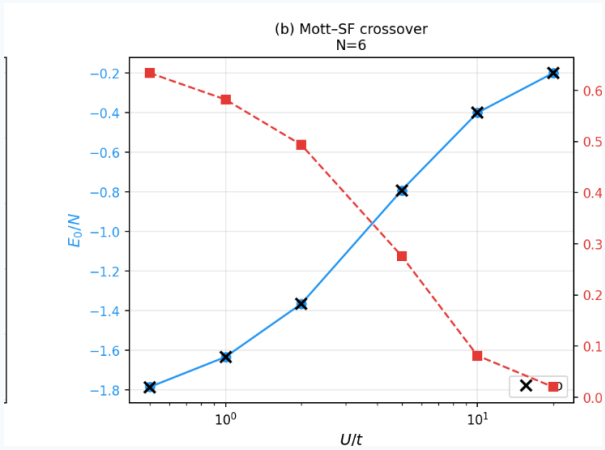
1D crossover accuracy | U(1) symmetry reduction | 2D-ready workflow

Bose-Hubbard ground states with exact U(1) sector projection

Sector projection is both a speedup and a correctness condition. By restricting VMC to the fixed-particle-number sector, the pipeline captures the Mott-superfluid crossover with sub-0.06% relative energy error across the 1D sweep and reaches the 3x3 2D target geometry at 0.1% relative error. The monotonic suppression of density fluctuations tracks the expected crossover physics cleanly.

Regime	Relative error	var(n)	Key result
Superfluid (U/t <= 2)	<= 5.6e-5	0.58-0.63	Large number fluctuations
Mott (U/t >= 5)	<= 6.0e-4	0.02-0.08	Frozen occupation
2D 3x3 (U/t = 10)	0.10%	0.313	Target geometry

Mott-superfluid crossover panel



The central Bose-Hubbard panel shows the crossover directly: energy per site rises with U/t while density fluctuations collapse toward the Mott regime. The same summary also validates n_max = 3 truncation and a 3x3 2D extension.

1.74e-4	<0.06%	0.1%	36.7x
ARNN rel. error (6x6 TFIM)	BH energy (1D sweep)	2D BH (3x3)	Max sector reduction

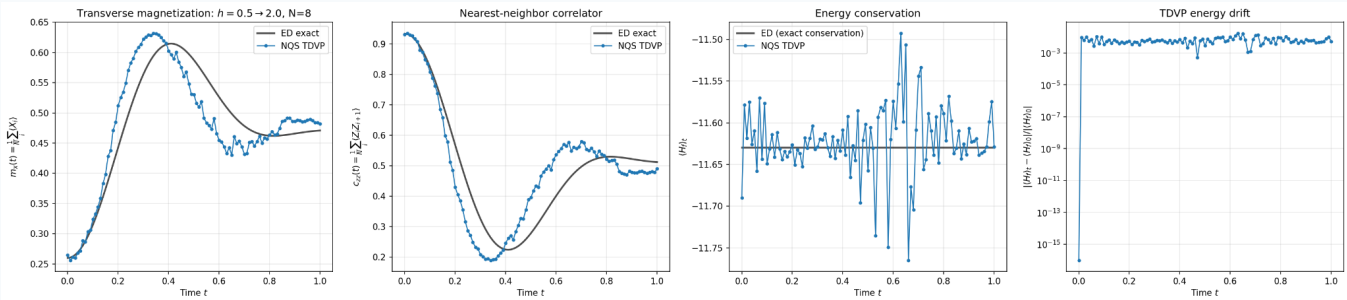
Real-Time Dynamics and Hybrid Pretraining

TFIM quench | symmetry enforcement | snapshot-to-VMC warm start

TDVP quench dynamics (N=8 TFIM, h: 0.5 -> 2.0)

Translation symmetry improves observable accuracy, and adding Z2 parity removes the dominant symmetry pathology: the spurious longitudinal magnetisation falls from 0.383 to 0.033, a 91% reduction. Energy MAE also improves, but the key interpretation is subtler: observable accuracy is essentially unchanged by the Z2 projection, so the remaining error floor is due to RBM expressivity plus Monte Carlo noise, not a residual symmetry bug.

Final TDVP run (translation + Z2)

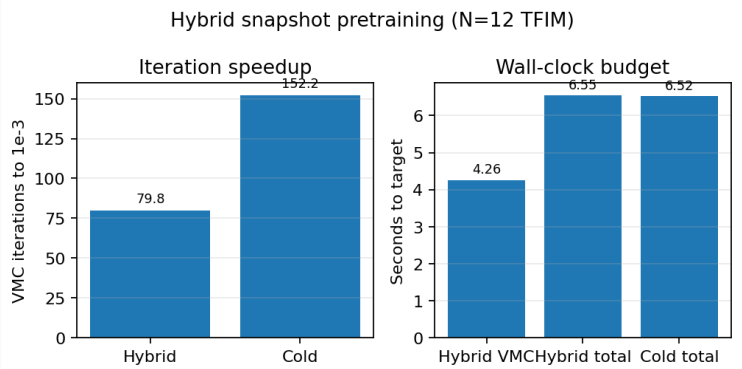


The final four-panel diagnostic confirms the intended outcome: correct qualitative oscillation structure, stable TDVP evolution, and strong suppression of the unphysical m_z signal after parity projection.

Hybrid snapshot pretraining (N=12 TFIM)

Born-sampled snapshots provide a reliable amplitude warm start for VMC. Across five seeds, the hybrid pipeline reaches the 1e-3 target in 79.8 iterations on average versus 152.2 for cold start, and it is faster by this metric in all five seeds. The important qualification is that the wall-clock comparison is roughly break-even at N=12 once the 2.29 s pretraining overhead is included.

Hybrid vs cold-start benchmark



Left: the robust result is a 1.91x reduction in VMC iterations to target. Right: the total wall-clock time is nearly unchanged at this size because pretraining overhead offsets the saved VMC iterations.

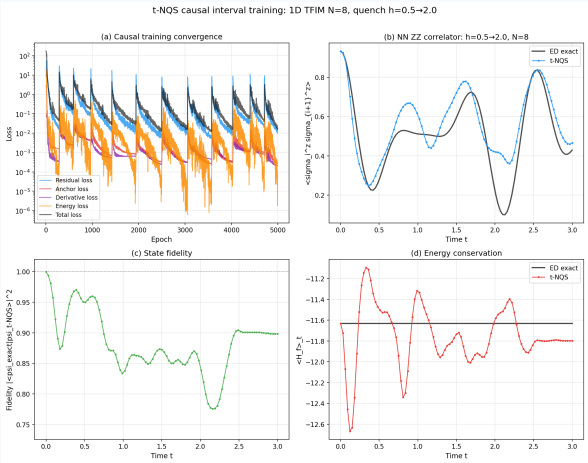
Time-Dependent NQS and Portfolio Synthesis

Causal interval training | fidelity-conservation tradeoff | next-step roadmap

t-NQS interval training (N=8 TFIM quench, T = 3.0)

The from-scratch PyTorch t-NQS implementation reproduces the key interval-training idea qualitatively: one parameter set is optimised over a time interval using a causal loss. Across the three runs, the best interpretation is not monotonic improvement but a Pareto tradeoff. Run 2 reaches strong conservation at moderate fidelity, while Run 3 pushes mean fidelity to about 0.88 at the cost of much larger energy drift. That tradeoff is consistent with the remaining architectural gap versus the paper's autoregressive attention-based transformer.

Best-fidelity t-NQS run



The final t-NQS figure makes the tradeoff visible: improved state fidelity and qualitatively correct ZZ dynamics, but noticeably weaker energy conservation than the more conservative Run 2 configuration.

Synthesis and Outlook

Together, these six summaries form a coherent NQS programme. The ground-state track moves from exact checks to a decisive 2D scaling advantage and then into Bose-Hubbard sector-projected workflows. The dynamics track separates symmetry fixes from ansatz limitations, shows where hybrid pretraining is already operationally useful, and maps the present t-NQS bottleneck to architecture rather than to the interval-training idea itself. The common thread is symmetry-aware model design plus carefully staged validation.

0.88	1.91x	0.1%
t-NQS mean fidelity	Hybrid speedup (iterations)	2D BH ground state